

Steven Haener

Atlanta, Georgia • stevenhaener98@gmail.com • 208-407-8019 • <https://stevenhaener.github.io/>

MS Robotics graduate (4.0 GPA) with 2+ years in the mechanical engineering industry and 4 years of robotics research. Experienced in robotics software/hardware development, focused on deployable systems that solve real-world problems.

EDUCATION

Georgia Institute of Technology Atlanta, Georgia
Master of Science in Robotics (4.0 GPA) May 2026

Selected Coursework: Robotics Research (ROS2), AI, ML, Deep Learning, Deep RL, Networked Control, Bio-mechatronics, Advanced Dynamics, Nonlinear Controls, Robotic Dynamics/Control

University of Idaho Moscow, Idaho
Bachelor of Science in Mechanical Engineering (3.76 GPA) May 2021

Minors in Math, Music, and German | Grand Challenge Scholars Program (GCSP) founding member

TECHNICAL SKILLS

Languages/packages: Python, C++, MATLAB, ROS2, IsaacSim, MuJoCo, PyTorch, Git, Linux, Bash, Numpy, Gazebo
Robotics: Navigation, State Estimation, Factor Graphs, Kinematics, Dynamics, CV, RL, Controls, YOLO
Design/manufacturing: SolidWorks, Fusion, AutoCAD, FEA, Plasma Cutting, 3D printing, CNC, Soldering, Welding

PROGRAMMING AND HARDWARE EXPERIENCE

Ground Control Robotics May 2026 – Aug 2026 | Atlanta, Georgia

- Migrating comms/controls to ROS2 to enable robot collaboration, sensor integration, and larger data-streams.
- Developing an Android-based controller for viewing on-board camera streams and adjusting gait parameters.
- Developing building-mapping strategies from on-board cameras and LiDAR using Gaussian Splatting techniques.
- Implementing RL policies in IsaacSim for autonomous exploration and traversal of unknown areas.

Dynamic Mobility Lab (Dynamo) July 2025 – Present | Atlanta, Georgia

- Published a conference paper (IROS) on GRF influence in motion retargeting for humanoid trajectories.
- Investigated the influence of introducing ground reaction force (GRF) data into the training of bipedal locomotion policies using motion capture data and reinforcement learning pipelines. We found that including GRF in the locomotion policy led to an increase in sample efficiency of ~10% and stabler gait for humanoids.
- Implementing policies in IsaacSim and MuJoCo for sim-to-sim and sim-to-real transfer on Unitree G1 hardware.

Georgia Tech Robotarium May 2025 – May 2026 | Atlanta, Georgia

- Migrated backend communication between ~20 robots on the testbed with the server/tracking cameras to ROS2-based client to better handle juggling synchronized tracking, real-time pose streams, and multi-robot coordination.
- Led to higher reliability, simpler sensor fusion, and higher bandwidth for large sensor data like cameras.
- Upgraded current robot hardware and implemented MATLAB code to include distance sensors (SLAM) and IMUs to enhance the tools provided to researchers that wish to run experiments with a larger arsenal of sensors.

Lab for Intelligent Decision and Autonomous Robots (LIDAR) Aug 2024 – Present | Atlanta, Georgia

- Designed and developed a foot in Fusion for a bipedal Cassie robot aimed at improving walking stability, slower walking gaits, and reduced actuation effort, leading to a longer lasting robot and more natural-looking movement.
- Integrated tactile/IMU/auditory sensors using C++ with actuated tarsal segments on both sides of the foot.
- Developed Python algorithm to leverage these additional signals for classification of the terrain being walked on, informing our control system of the terrain characteristics so it can tune its strategy for increased stability.
- Writing a journal paper describing design/testing methodologies and results.

INDUSTRY/WORK EXPERIENCE

Georgia Tech | *Graduate Teaching Assistant* May 2025 – Present | Atlanta, Georgia
– Held office hours, wrote exams, and graded papers for graduate-level AI and Machine Learning courses.

Campos EPC Inc. | *Mechanical Engineer I* June 2023 – June 2024 | Salt Lake City, Utah
– Assessed and proctored IFC drawings and surveyed pipelines, automation systems, pumps, and control valves.

Partner Engineering and Science Inc. | *Project Engineer* June 2022 – June 2023 | Salt Lake City, Utah
– Verified conditions of mechanical systems in commercial spaces and prepared technical reports on findings.

Terraclear Inc. | *Robotics Engineer* Nov 2021 – Mar 2022 | Bellevue, Washington
– Implemented FEA and design analysis in SolidWorks to reduce the weight of the rock picker by nearly 40%.